Latitudinal effects on weight and age dependent survival of Red fox *Vulpes vulpes*

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Abstract

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Introduction

Age dependent survival is central to understand population dynamics and life-history evolution (Lande et al., 2003; Stearns, 2000). Patterns of age dependent survival are species specific (Loison et al., n.d.; Martin, 1995), and determine resource allocation strategies as body size and morphology (Brooks et al., 2016; Roff, 1986; Sibly and Brown, 2007). Life-history traits show positive covariation that results in a slow-fast continuum of population dynamics (Oli, 2004), and is closely correlated to generation time (Kraus et al., 2005). Slow species are long-lived with low-reproductive rate, whereas fast species are short lived that allocate resources to early maturation and high fertility. The effect of stochastic processes is higher at the fast end of the continuum (Sæther et al., 2013). Resource allocation strategies determine body size and affect central traits in life-history evolution as reproduction and survival (Brooks et al., 2016; Roff, 1986; Sibly and Brown, 2007).

Species with a wide geographical distribution are exposed to a broad range of and climate conditions, which potentially could result in different optimal values of life history-traits. For example, Bergmann's rule states that there is a positive correlation of weight of a species with latitude inhabited (Blanckenhorn, 2000). However, as long as the different sub-populations are connected, evolution should favor adaptive phenotypic plasticity (Kawecki and Stearns, 1993; McNamara and Houston, 1996). The Red fox *Vulpes vulpes* is a highly adaptive species, and one of the most widely distributed carnivores. It can utilize a wide range prey from voles to roe deer (Englund, 1970; Kjellander and Nordström, 2003), and also feed on vegetables and human waste (Carvalho and Gomes, 2001). It benefits from human settlements and agricultural activities (Gallant et al., 2020; Jahren et al., 2020; Vuorisalo et al., 2014), but is limited by deep snow and harsh winters at northern latitudes (Bartoń and Zalewski, 2007; Frafjord and Stevy, 1998). Demographic data and other life-history characteristics place the Red fox on the fast side of the fast-slow continuum (Heppell et al., 2000). Females start to reproduce at an age about 10 months (Englund, 1970), and survival estimates are low both for adults and subadults (Devenish-Nelson et al., 2013; Willebrand et al., 2022). Despite extensive mortality during the first outbreaks of sarcoptic mange, red fox populations appeared to recover within decades after the outbreak(Soulsbury et al., 2007; Willebrand et al., 2022). Devenish-Nelson et al. (2013) compared demographic data of eight different fox populations in Australia, UK, Sweden and USA, and concluded that the population dynamics ranged medium to fast life history strategies.

Red fox is distributed throughout all of Sweden's 450,295 km^2, which stretches from from 55.38° to 67.86° latitude. The northern parts are sparsely populated and have a subarctic climate, and the area north of the polar circle (66.34°) is characterized as arctic tundra. The southern parts, on the other hand, have an animal diversity and human densities similar to southern European countrie, and winters only have shorter periods of snow and sub-zero temperatures. The aim with this study was to evaluate the effects of a latitudinal gradient on age related body size of males and females and age-dependent survival patterns of ered foxes in Sweden. We analyze age, sex, and weight from 6 022 harvested red foxes between 1967 and 1971, and expect body size and age dependent survival to increase with latitude.

Methods

Collecting harvested foxes

The data used in this analysis is part of a monitoring program that was initiated by the late professor Englund to investigate age, reproduction and morphological characteristics of the Red fox. In this study, we used 6 022 harvested foxes were collected from hunters between 1967 and 1971 when efforts were made to retrieve foxes from all parts of Sweden. Carcass weight was recorded to the nearest hectogram, and age was determined by microscopic examination of cementum layers of the canine teeth. See (Englund, 1965) and Englund, 1970 for additional details.

Latitudinal regions. We grouped the collected foxes in four latitudinal regions similar to Englund, 1970. See (*Geography of Sweden* 2025; *Markanvändningen i Sverige =* 2013; Zheng et al., 2004) for more details. Sweden covers a wide range of climatic zones, and stretches more than 1 500 km from south to north.

N:: *1461 males and 1128 females.* The northernmost part of Sweden that stretches mostly north of 62° latitude to north of the polar circle. The region is characterized by low productivity and a net primary production (NPP) below 300. The region is dominated by boreal forests (68%) and open tundra (27%), and experience long-winters and deep snow cover. Microtines, hares and grouse are common prey speciesfor red foxes. Human waste, slaughter remains and carcasses from domestic reindeer *Rangifer tarandus* and moose *Alces alces* are occasionally available in autumn and winter. The region has less than 6 inhabitants per km^2.

NC:: *553 males and 437 females.* North-Central Sweden, between 60° and 62°latitudes, is dominated by boreal conifer forest (84%) interspersed with agricultural (4%) areas. The nortwest parts contain open tundra (7%). Winters vary, and snow cover usually stays from November until March. Microtines, hares and grouse but also roe deer are among important prey species. Human waste, slaughter remains and carcasses from domestic reindeer and moose are occasionally available in autumn and winter. Human settlements are more common than in NC, and the region has about 12 inhabitants per km^2.

SC:: *929 males and 809 females.* South-Central Sweden, between 58° and 60°latitudes is characterized by forests (63%) and agriculture (22%) areas. Broad leaf tree species are common in forests. Winters are short with low amounts of snow. Prey species are more diverse than in NC but microtines are still important. Human waste and slaughter remains from ungulate species including wild boar *Sus scrofa* are available in autumn and winter. The region contains most of the Swedish human population with close to 100 inhabitants per km^2, and 6% is classified as urban areas.

S:: *374 males and 289 females.* Southern part of Sweden, which is below 58° latitude. The region contains forest (65%) and agriculture (19%) areas, and forests contain several different broad leave species in addition to conifers. It is dominated by short winters and a climate similar to central Europe. Prey is abundant, including rabbits *Oryctolagus cuniculus* that are not present in regions further north(Erlinge et al., 1984). Human waste and slaughter remains are available occasionally in autumn and winter. Inhabitants are close to 100 per km^2 and 7% is classified as urban areas.

Analysis

We evaluated the effect of area, sex and age on carcass weight, and used a general linear model assuming a normal distribution with a unity link. We separately analyzed the effect of area and sex for adults and sub-adults because we assumed that weight gain was higher for sub-adults compared adults. Area and sex were treated as categorical variables whereas age of adults was considered as a continuous variable. We used residual diagnostics to validate models with the package DHARMa (Hartig, 2022).

Age-at-harvest data are usually updated for many wildlife species and can be used to supplement survival estimates from mark-recapture and known-fate studies. Life-table analysis based on age at harvest data is an indirect method to estimate survival, and its use is restricted by assumptions as of a stable age structure, constant survival, and a population at equilibrium (λ= 1). These assumptions have been evaluated and to different degrees relaxed (Udevitz and Ballachey, 1998). and a recent study by Skelly et al. (2023) proposed a Bayesian approach to estimate survival probability from age-at-harvest data when auxiliary data to validate assumptions are not available. The model include flexible definitions of the likelihood and can account for uncertainty in age structure, population growth rate, and harvest selectivity by assigning prior distributions. In addition, survival probability may vary as a function of environmental variables.

Here we have used the outlined Bayesian model proposed by Skelly et al. (2023). Our data contained age classes from first year sub-adults (age-class 1) to 12 years of age. We pooled all ages above 8 years to age class 9, and tabulated data by age, sex and region. We did not include any parameter on the relative selectivity to be harvested. We used harvest statistics from counties in region N and NC to estimate indices of population change, and used the first and second quartile (*dunif(0.8153, 1.1000)*) as the prior for λin the model. We developed 8 models with different combinations of sex, age and area affecting survival probabilities (table 3 lists the different models). We evaluated model fit with posterior predictive checks (Kéry, 2010), and used deviance information criterion (*DIC*) for model selection. All models were developed in a Bayesian framework using the JAGS-language (Plummer et al., 2003) (see appendix). All data preparation and analysis were made in R (R Core Team, 2024).

Results

Weight decrease with latitudes

Carcass weights of red foxes decreased from the southernmost to the northernmost region. Figure 1 show the weights in the different areas for ages 1-4 years separated by sex. In adults, the difference in average weight between the southernmost and northernmost regions was larger (1.64 kg) than the average difference (1.21 kg) between males and females. Similarly in sub-adults, the average difference in weight (1.47 kg) between North and South was also larger than the average difference (0.98 kg) between males and females. See table 1 & table 2. Adults also increased in weight with age, but only 69 g on average per year.

Age dependent survival increase with latitude

The full model, including the age, sex and region, was associated with the lowest DICs, and was 2.9 units lower than the next best model including only sex and age. See table 3. The bayesian p-value was high (0.712) for the full model, whereas the bayesian p-value of the model with only age and sex (0.54) was closer to the preferred 0.5. Here we present the results of the

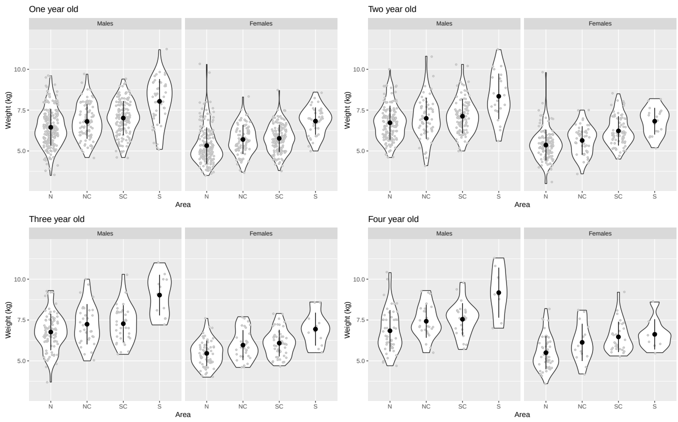

Figure 1 – Comparison of carcass weights of adult foxes aged 1-4 years in a latitudinal gradient of four regions in Sweden.

Table 1 – Evaluation of variation in weight dependent on sex and region on adult red foxes in Sweden. A generalized linear model with family Gaussian and unity link was used to model the variation. Numbers in parenthesis are standard error of the parameter.

	Dependent variable:
	Weight (kg)
Age	0.069***
	(0.014)
North Central	0.420***
	(0.060)
South Central	0.583***
	(0.052)
South	1.638***
	(0.085)
Female	−1.211***
	(0.044)
Constant	6.446***
	(0.049)
Observations	2,340
Log Likelihood	−3,442
Akaike Inf. Crit.	6,897
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 2 – Evaluation of variation in weight dependent on sex and region for sub-adult red foxes in Sweden. A generalized linear model with family Gaussian and unity link was used to model the variation. Numbers in parenthesis are standard error of the parameter.

	Dependent variable:
	Weight (kg)
North Central	0.209***
	(0.055)
South Central	0.392***
	(0.045)
South	1.471***
	(0.068)
Sex	−0.980***
	(0.039)
Constant	5.931***
	(0.032)
Observations	2,733
Log Likelihood	−3,863
Akaike Inf. Crit.	7,736
Note:	*p<0.1; **p<0.05; ***p<0.01

full model because of the low DIC, and our aim to evaluate regional differences. The parameter values for the logistic regression is presented in table 4, and the estimated age dependent survival for the different areas are presented separately for males and females in figure 2. The estimated age-dependent survival show a substantial lower survival of sub-adult foxes compared to adults. Adult survival was stable until the age of eight year when it decreased dramatically. Average survival was lower in the two southern areas compared to the two northern areas. Thus, regional survival showed an opposite pattern to regional differences in weight, although the survival in region NC appeared higher than in N.

The relative age distribution in the different areas for each sex is presented in figure 3. The full model resulted in an overall growth rate of λ= 1.040 (95% Credible Interval, 0.898 - 1.098).

Table 3 – Model selection and Bayesian p-values for estimates of age-dependent survival of red foxes in Sweden including different sets of parameters.

Model	Bayesian p-value	DIC
Age + Sex + Region	0.712	369.235
Age + Sex	0.544	372.144
Age + Region	0.025	400.950
Age	0.012	402.833
Sex + Region	0.000	496.588
Sex	0.000	501.042
Region	0.000	532.792
Intercept	0.000	536.054

Table 4 – Parameter estimates of age-dependent survival including age, sex and region as variables. Note that parameters are expressed on the logistic scale. See appendix for model details.

Parameter	Mean	Lower 2.5%	Upper 97.5%
Intercept	-0.560	-0.791	-0.396
Age 2	0.937	0.624	1.285
Age 3	0.694	0.399	1.022
Age 4	1.178	0.704	1.799
Age 5	0.747	0.270	1.325
Age 6	1.067	0.360	2.092
Age 7	1.034	0.173	2.424
Age 8	-0.017	-0.502	0.463
Age 9	0.215	0.138	0.293
Region NC	0.092	-0.011	0.195
Region SC	-0.053	-0.140	0.032
Region S	-0.082	-0.204	0.042

Discussion

Our results show that red foxes decrese in weight with increased latitude and decreased productivity are similar to Frafjord and Stevy (1998), and do not suppport Bergmann's rule that variation in body size can be explained by the need to conserve heat in colder climates. Several authors have critizied Bergmann's rule for being to simplistic, and have propsed other mechanisms that explain body size variation within and between species (Blackburn et al., 1999; McNab, 1971). The availability of nutrients during growth is important for body size (Geist, 1987), and can be explained by density dependent competition for food. It is well known that density dependent affect body size in high density deer populations (Gaillard et al., 2003). Cavallini (1995) reported that Red fox males, but not females, were heavier in the northern than in the southern

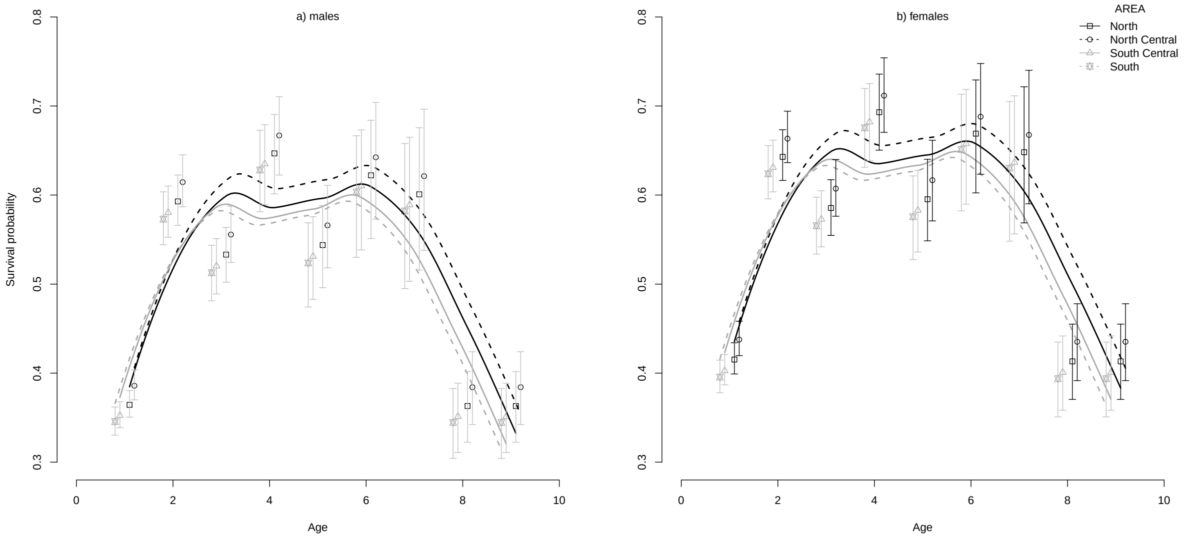

Figure 2 – Estimated age dependent survival of male and female red foxes in four different areas along a latitudinal gradient in Sweden. Note that a small deviation has been added to age to improve the readability of different regions.

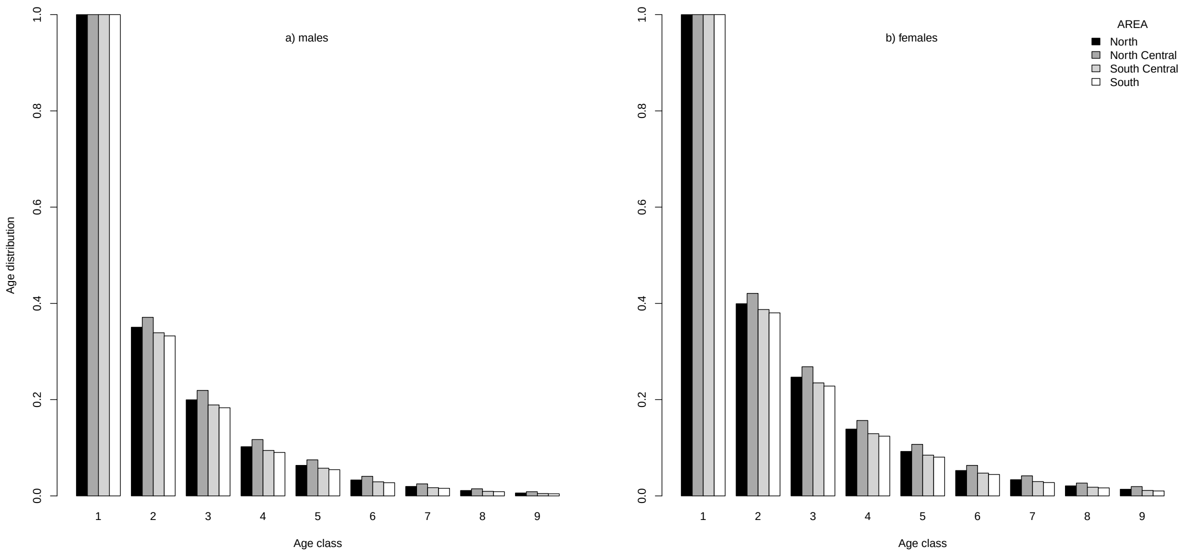

Figure 3 – Estimated relative age distribution of male and female red fox in four different regions along a latitudinal gradient in Sweden. The age distribution of age 1 is set to 1.

part of a study area in Italy, and expected that weight and size of the Red fox should increase with latitude, and emphasized the importance of negative density dependence for variation in body sizeCavallini (1995). However, density-independent fluctuations in the environment contributed to body size variation of female brown bears in addition to density-dependent factors (Zedrosser et al., 2006), the these area differences in productivity (Hjeljord and Histøl, 1999) Our results suggest that individual and positively to .

Growth during the natal period These findings are rare experimental evidence that prenatal and postnatal investments have distinct effects in moulding individual life history and fitness in wild mammals. (Vitikainen et al., 2023). Our results suggest that mass gain during lactation, possibly but not necessarily related to the amount of maternal care received, affects adult mass and reproductive success. (Albon et al., 1987; Festa-Bianchet et al., 2000)

Prey size. Several authors have pointed out the relationship between carnivore body size and the usual weight of their prey (Rosenzweig 1966; Schoener 1969a; Kruuk 1972, 1975; Schaller 1972; Kleiman and Eisenberg 1973). (Gittleman, 1985).

SURVIVAL, age distribution sex ratio,

The estimates of age-dependent survival are similar to earlier studies

Survival values Survival and latitude. Density and reproduction.

South Higher density and dispersal northwest. Density dependent mortality.

Assumptions: Pre mange population high and stable.

Variation in recruitment. Reproduction and dispersal. Population dynamics of large herbivores: variable recruitment with constant adult survival Gaillard

Age structure (Yoneda and Maekawa, 1982)

Immigration to N and NC from SC and S would bias the age dependent survial high. SMall foxes are pushed north?

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Appendix

Jags models

Model use to estimate age-dependent survival:

```
model{

  for(k in 1:K){
    b[k] ~ dlogis(0, 1) # logistic prior
  }

 lambda ~ dunif(0.8153, 1.1000)

### CALCULATING STABLE STAGE DISTRIBUTION

for(i in 1:G){  # looping over all groups
  # survival probability
    logit(p[i, 1]) <- b[1] + b[9] * sex[i] +b[10] * area3[i] + b[11] * area4[i]
# survival covariates = sex + year
    g[i, 1] <- p[i, 1]  # survive to next stage class
    z[i, 1] <- 0  # remain in current stage class
    w[i, 1] <- 1  # proportion to stable stage

 for(s in 2:(S - 1)){  # loop through all stage classes
      logit(p[i, s]) <- b[1]  + b[s] + b[9] * sex[i] +b[10] * area3[i] + b[11] *
# survival for each stage class + survival covariates = sex + year
      g[i, s] <- p[i, s]
      z[i, s] <- 0
      w[i, s] <- g[i, s - 1] / (lambda - z[i, s]) * w[i, s - 1]
    }

 logit(p[i, S]) <- b[1] +  b[S - 1] + b[9] * sex[i] +b[10] * area3[i] + b[11] *
# survival for each stage class + survival covariates = sex + year
    g[i, S] <- 0
    z[i, S] <- p[i, S]
    w[i, S] <- g[i, S - 1] / (lambda - z[i, S]) * w[i, S - 1]

# stable stage distribution
    C[i, 1:S] <- w[i, 1:S] / sum(w[i, 1:S])

### EXPECTED COUNT IN EACH STAGE CLASS

  ey[i, 2] <- p[i, 1] * C[i, 1]
  ey[i, 3] <- p[i, 2] * C[i, 2]
  ey[i, 4] <- p[i, 3] * C[i, 3]
  ey[i, 5] <- p[i, 4] * C[i, 4]
  ey[i, 6] <- p[i, 5] * C[i, 5]
  ey[i, 7] <- p[i, 6] * C[i, 6]
  ey[i, 8] <- p[i, 7] * C[i, 7]
  ey[i, 9] <- p[i, 8] * C[i, 8] + p[i, 9] * C[i, 9]
  ey[i, 1] <- lambda - sum(ey[i, 2:9])
```

```
### LIKELIHOOD
    y[i, 1:S] ~ dmulti(ey[i, 1:S], N[i])

### MODEL CHECKING
    y_new[i, 1:S] ~ dmulti(ey[i, 1:S], N[i])

  for(n in 1:S){
      g_obs[i, n] <- y[i, n] *
      log(y[i, n] / (N[i] * ey[i, n] / sum(ey[i, 1:S])))
      g_new[i, n] <- y_new[i, n] *
      log(y_new[i, n] / (N[i] * ey[i, n] / sum(ey[i, 1:S])))
    }
  }

  cs_obs <- 2 * sum(g_obs)
  cs_new <- 2 * sum(g_new)
  # bayesian p-value
  bp <- step(cs_new - cs_obs)
}
```

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